



DoubleML

Closing

DoubleML Workshop

February 27 & March 06, 2026

Economic AI

Congratulations!

 You completed the **2-Days Intensive Training in Causal ML with DoubleML**! 

Day 1

- Basics of causal inference
- Challenges to Causal ML
- **DoubleML**: An overview
- Multiple treatment levels and treatment effect heterogeneity

Day 2

- Recap on Double Machine Learning
- **DoubleML** in Practice
- Difference-in-Differences with **DoubleML**
- Sensitivity Analysis

What's next?

- Now, you are ready to kick-off your own causal ML projects 
- Some references that might help you
 - Chernozhukov et al. (2024)
 - Huber (2023)
 - Cunningham (2021)
 - Facure and Germano (2021)
 - Ruiz de Villa (2023)
- Try to find the right DML methods for your problems

Thank you very much!

- We really appreciate your feedback!
- Keep in touch and stay tuned for upcoming courses and events
- Spread the word 😊
- Contact me with any questions related to industry use cases or research projects
philipp.bach@fu-berlin.de
- You will receive a link with the teaching materials

References

References

- Bach, Philipp, Victor Chernozhukov, Malte S Kurz, and Martin Spindler. 2022. “DoubleML—an Object-Oriented Implementation of Double Machine Learning in Python.” *Journal of Machine Learning Research* 23: 53–51.
- Bach, Philipp, Malte S. Kurz, Victor Chernozhukov, Martin Spindler, and Sven Klaassen. 2024. “DoubleML: An Object-Oriented Implementation of Double Machine Learning in R.” *Journal of Statistical Software*. <https://doi.org/10.18637/jss.v108.i03>.
- Chernozhukov, Victor, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis. 2024. *Applied Causal Inference Powered by Machine Learning and AI*. CausalML-book.org. <https://causalml-book.org>.
- Cunningham, Scott. 2021. “Causal Inference.” In *Causal Inference*. Yale University Press.
- Facure, Matheus, and Michell Germano. 2021. “matheusfacure/python-causality-handbook: First Edition.” Zenodo. <https://doi.org/10.5281/zenodo.4445778>.
- Huber, Martin. 2023. *Causal Analysis: Impact Evaluation and Causal Machine Learning with Applications in R*. MIT Press.
- Ruiz de Villa, Alex. 2023. *CAUSAL INFERENCE FOR DATA SCIENCE*. Manning.